TRUSTLENS AI

Human-Governed Machine Learning for Reliability-Aware Risk Classification

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Research question. Can calibrated prediction, drift and out-of-distribution checks, explainability and explicit human-review rules expose unreliable model outputs more effectively than confidence scores alone?

Why this project matters

High-stakes models can appear confident while operating outside their evidence base. TrustLens treats uncertainty, monitoring and escalation as executable system behaviour rather than documentation added after model development.

Research design

The case study uses 1,000 historical South German Credit records. A stratified 20% test partition was reserved before development. Logistic regression, random forest and gradient-boosting candidates were evaluated with five-fold cross-validation, asymmetric error cost and calibration measures. The selected cost-sensitive random forest was sigmoid-calibrated, used a threshold fixed on development evidence and excluded two sensitive or ambiguous inputs. It was evaluated once on the untouched 200-record partition.

Balanced accuracy	Higher-risk recall	Precision	Weighted cost	Automated tests
0.656	**0.833**	**0.407**	**123 vs 300**	**27 passing**

Governance and reliability evidence

The prototype implements probability calibration, population-drift detection, individual out-of-distribution screening, local sensitivity explanations, subgroup diagnostics and pause-or-review rules. A 20% development review budget captured 23% of observed errors. Post-hoc 95% Wilson intervals report higher-risk recall as 0.720-0.907 and precision as 0.324-0.495.

Failure analysis and limitations

The model produced 73 false positives and 10 false negatives. Error-slice diagnostics found sharply different failure rates across recorded checking-account-status codes, reinforcing the need for human review. These results are descriptive, not causal: the dataset is small, dates from 1973-1975, contains limited demographic information and has no external validation. TrustLens is an educational research prototype and must not make real lending or eligibility decisions.

Doctoral research direction

A doctoral extension would test governance-aware model selection on contemporary multi-institutional data, evaluate selective prediction and abstention, study realistic temporal drift, and measure whether human reviewers can correct errors without automation bias. The broader goal is responsible AI that expands access to technology while making uncertainty and system limits visible.

Repository	https://github.com/akshith2001/trustlens-ai
Live dashboard	https://trustlens-governance-ai.streamlit.app/

Research portfolio summary | 22 August 2026